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EDUCATION

Northeastern University

Master of Science, Data Analytics Engineering GPA: 3.95/4.0

Boston, MA

Expected December 2026

Coursework: Machine Learning Operations; Deep Learning for AI; Data Mining in Engineering.

VIT Bhopal University

Bachelor of Technology, Computer Science Engineering GPA: 3.55/4.0

Bhopal, India

May 2024

Publication: Co-author, peer-reviewed article on ML for hyperspectral image classification, *Multimedia Tools and Applications* (Springer), 2025.

EXPERIENCE

GPT Integrators

Atlanta, GA (remote)

AI Engineer Intern

August 2025 – January 2026

- Eliminated roughly 90% of manual data entry for 10+ enterprise clients, processing 40,000+ records per month, by building an extraction service integrating the OpenAI API behind FastAPI endpoints.
- Raised extraction accuracy from 95% to 98% across that monthly volume by designing statistical validation benchmarks and running systematic error analysis to root-cause recurring failure modes.
- Cut issue-resolution time 60% for 5+ logistics clients by embedding with client stakeholders, analyzing live chatbot logs, redesigning conversation flows, and presenting trade-offs to non-technical decision-makers.

Tally Solutions

Bangalore, India

Software Development Engineer Intern

June 2023 – July 2023

- Enabled cross-functional PMO reporting by architecting a PostgreSQL warehouse schema handling 50+ concurrent datasets.
- Cut dashboard load time 40% across 10+ reports by constructing ETL pipelines and refactoring SQL queries with targeted indexing.

PROJECTS

Aurora — Customer Operations Agent (GitHub)

2026

- Resolved customer-operations tickets end-to-end with an action-taking LLM agent on LangGraph (intake → plan → act → resolve) calling 16 schema-validated tools, grounded by Chroma RAG and episodic/semantic memory.
- Held zero forbidden actions across 46 standard and adversarial scenarios (scripted model) with a deterministic policy engine and layered guardrails: injection sanitization, policy-gated writes, loop breakers, cost caps, and reply checks (grounded, own-customer records only); repeated weaker-model runs surfaced a refund-retry gap, closed with a replay test.
- Engineered the action path for production reliability — exactly-once execution under duplicate delivery and worker restart (idempotency keys, Redis/RQ durable queue, retries, dead-letter), benchmarked at $3.7\times$ throughput on 4 workers — with a CI eval harness (task success, tool-trajectory precision/recall, bias-validated LLM-as-judge; 72 tests), an async FastAPI service, and an MCP server exposing the same policy-gated tools.

F1 Race Intelligence Agent (GitHub)

2026

- Made an agent's tool definitions 85% smaller (13,487 → 2,014 tokens, measured with LangChain serialization and tiktoken) by consolidating 79 tools into 8 intent-level dispatchers with guarded read-only text-to-SQL; coverage tests confirm all 79 still work.
- Architected the LangGraph analytics agent over seven seasons of F1 telemetry (2018–2024) on TimescaleDB, Neo4j and Qdrant, behind FastAPI with a multi-provider LLM router.

Resolv — Entity Resolution System (team project)

Spring 2026

- Built the inference pipeline for a 6-person entity-resolution system: candidate generation via 7-key blocking, transformer semantic scoring, and deterministic field-rule vetoes, feeding graph-based network-risk scoring for Customer 360, KYC/OFAC screening, and fraud detection.
- Shipped the FastAPI + React console and the Cloud Run deployment, and contributed to the GCP MLOps stack (Vertex AI, Airflow, Terraform) with Kolmogorov–Smirnov drift monitoring that triggers retraining.

Additional Projects

DataWash — authored and published to PyPI (`pip install datawash`) an open-source data-quality library: PII detection (6 categories), profiling, cleaning, and reproducible cleaning-code generation; 200+ tests, ~92% coverage. **E-Commerce Experimentation** — A/B-testing methodology (power analysis, bootstrap CI) on a simulated free-shipping treatment over the UCI Online Retail dataset; XGBoost + SHAP churn model at ROC-AUC ~0.81 after removing target leakage.

TECHNICAL SKILLS

Languages, Data & Statistics: Python, SQL, TypeScript/JavaScript, PySpark, Pandas, NumPy, Git, PyTorch, TensorFlow, Scikit-learn, XGBoost, LoRA/PEFT, Fine-Tuning, A/B Testing, SHAP, MLflow

LLM & Agentic AI: LangChain, LangGraph, RAG, AI Agents, Tool Use/Function Calling, Structured Outputs, Guardrails, Human-in-the-Loop, Prompt & Context Engineering, LLM Evaluation, LLM-as-a-Judge, Model Context Protocol (MCP), HuggingFace, ChromaDB, Qdrant, OpenAI & Anthropic Claude APIs

Infrastructure & MLOps: AWS, GCP (Vertex AI, Cloud Run), Airflow, Terraform, Docker, Redis, GitHub Actions/CI, FastAPI, REST APIs, React, PostgreSQL, Neo4j, Power BI